

Structural Approach

ECON 8120

Spring 2026

Prof. Josh Kinsler
University of Georgia

INTRODUCTION

- ▶ What steps are required to estimate a structural model?
 - ▶ walk through each step on an example model
- ▶ Issues we won't address....
 - ▶ the material we discuss today assumes you have data
 - ▶ and that you have sufficient understanding of your data
 - ▶ it also assumes you have an understanding of your preferred coding language
 - ▶ these are all non-trivial steps, but they are typically covered in other classes
 - ▶ we will continue to develop these skills throughout the course

STEPS TO PERFORMING STRUCTURAL ESTIMATION

- ▶ Mike Keane gave a **talk** at the University of Chicago in 2015 and listed these steps:

1. Theoretical Model Development
2. Practical Specification Issues
3. Solving the Model
4. Understanding How the Model Works
5. Estimation
6. Validation
7. Policy Experiments

AN EXAMPLE MODEL

- ▶ To help fix ideas, let's revisit a commonly used model in introductory econometrics:

$$\log(w_i) = \beta_0 + \beta_1 s_i + \beta_2 x_i + \beta_3 x_i^2 + \varepsilon_i \quad (1)$$

where we have cross-sectional data and

- ▶ i indexes individuals
 - ▶ w_i is employment income
 - ▶ s_i is years of schooling
 - ▶ x_i is years of work experience (or, more commonly, “potential” work experience)
 - ▶ ε_i is anything else that determines income
-
- ▶ We want to estimate $(\beta_1, \beta_2, \beta_3)$, which drive returns to human capital investment

QUICK REVIEW

- ▶ It is nearly certain that equation (1) suffers from omitted variable bias
 - ▶ i.e. there are lots of factors in ε_i that are correlated with both s_i and w_i
- ▶ Thus, our estimates of $(\beta_1, \beta_2, \beta_3)$ will not be causal
- ▶ We could try to get causal estimates using a variety of identification strategies:
 - ▶ find a valid instrument for s_i (Angrist & Krueger, 1991; Card, 1995)
 - ▶ exploit a discontinuity in s_i (Ost et al., 2018)
 - ▶ randomize s_i (Attanasio et al., 2011)
 - ▶ etc.

A STRUCTURAL VIEW OF EQUATION (1)

- ▶ We know that equation (1) will produce biased estimates, but why? Some possibilities:
- ▶ **ability bias**
 - ▶ s_i and w_i are both positively correlated with unobservable cognitive ability
- ▶ **comparative advantage**
 - ▶ multidimensional unobservable ability $\implies$ self-selection into schooling
- ▶ **credit constraints**
 - ▶ s_i is a costly investment; some people may not be able to borrow enough
- ▶ **preference heterogeneity**: differing tastes for s_i , differing discount rates
- ▶ **sample selection**: only observe wages for those that work

A STRUCTURAL APPROACH

- ▶ Consider a scenario where you lack experimental or quasi-experimental variation in s_i
- ▶ The sources of bias in equation (1) stem largely from the endogeneity of s_i
- ▶ A structural approach would be to jointly model the **choice** of s_i and labor supply along with wages
- ▶ Of course the availability of exogenous variation doesn't preclude the structural approach....and in fact will make it significantly more convincing

1. THEORETICAL MODEL DEVELOPMENT

- ▶ Since schooling has an up-front cost and long-term benefit, a dynamic model seems like the place to start
 - ▶ period 1: decide how much schooling to get
 - ▶ period 2: choose whether or not to work; if working, receive income by (1)
 - ▶ individuals choose schooling to maximize lifetime utility
- ▶ Preferences (denote utility in period t by u_t , with s , x and w defined previously)

$$\begin{aligned}u_1(z, c, \eta_1) &= f(z, c, \eta_1) \\ u_2(w(s, x), k, \eta_2) &= g(w(s, x), k, \eta_2)\end{aligned}\tag{2}$$

where z is family background, c is schooling costs, k is number of kids in adult household and η_t are unobservable preferences (similar to ε in equation (1))

1. THEORETICAL MODEL DEVELOPMENT

- ▶ With discount factor $\delta \in [0, 1]$, the discounted lifetime utility function is then

$$V = u_1(z, c, \eta_1) + \delta u_2(w(s, x), k, \eta_2) \quad (3)$$

- ▶ Equations (1)-(3) define our model
- ▶ This model is still highly unrealistic, but at least we have something
- ▶ A number of important questions arise (but we'll ignore these for today)
 - ▶ Where is cognitive ability? What exactly does c represent? Where are loans?
 - ▶ Maybe people should care about consumption in period 2, not income
 - ▶ Does family background really only enter u_1 and not $\log(w)$?
 - ▶ Should x in (1) be a function of s ? (Lower $s \implies$ longer working life)
 - ▶ What are people's beliefs about future k when deciding s ?

1. OVERVIEW OF THE THEORETICAL MODEL

- ▶ It takes knowledge and practice to write down even a simple model
 - ▶ need a detailed understanding of the subject and math/econ more generally
 - ▶ good practice to start from canonical models
- ▶ Key Components
 - ▶ **Exogenous variables:** family background (z); schooling costs (c); children in household (k)
 - ▶ **Endogenous variables:**
 - ▶ **Choices:** schooling (s); period-2 work decision
 - ▶ **Outcomes:** labor income (w)
 - ▶ **Unobservables:** income (ε); preferences (η_t)
 - ▶ **Model parameters:** returns to human capital (β); discount factor (δ); other parameters implied by $f(\cdot)$ and $g(\cdot)$

1. SOURCES OF IDENTIFICATION

- ▶ Joint distribution of unobservables
 - ▶ assume a joint distribution for $(\varepsilon, \eta_1, \eta_2)$
 - ▶ links schooling, employment, and wage outcomes
- ▶ Exogenous shifters with exclusion restrictions
 - ▶ z, c shift schooling utility but not wages directly
 - ▶ k shifts employment utility but not wages
 - ▶ these variables generate variation that disciplines choices without mechanically entering all equations
- ▶ Timing and economic structure
 - ▶ schooling chosen before wages and employment
 - ▶ work decision made given expected wages
 - ▶ dynamic trade-offs help identify preferences and discounting

Takeaway: Identification comes from *structure, exclusions, and distributional assumptions*

2. PRACTICAL SPECIFICATION ISSUES

- ▶ Now that we have a model, we need to figure out how to take it to data
- ▶ This is where we apply knowledge about our data and stats/econometrics
- ▶ Key data questions:
 - ▶ Which variables in the model are observed in our data set?
 - ▶ Are they reliably measured?
- ▶ Key specification questions:
 - ▶ How to model η_t and ε ? (Need to make distributional assumptions)
 - ▶ Functional forms of $f(\cdot)$ and $g(\cdot)$
 - ▶ Should s be continuous (years of schooling) or discrete (college/not)?

2. PRACTICAL SPECIFICATION ISSUES

- ▶ We won't get into too many details about this today, but specification is important!
- ▶ What determines the specification is often:
 - ▶ what is reliably measured in the data
 - ▶ what is computationally feasible to estimate
 - ▶ restrictions on the choice set, state space, time horizon, number of players, etc.
- ▶ Parameters of the model either need to be estimated or calibrated
 - ▶ e.g. often we don't have reliable data to allow us to estimate δ ; we must calibrate it
- ▶ Computational feasibility often governs how we specify the different functions
 - ▶ e.g. linear-in-parameters with additively separable unobservables (like equation (1))

EXAMPLE WITH REAL DATA

- ▶ Here is some real data from the a recent round of the NLSY97
 - ▶ We have demographics/background, wages, employment status, education, fertility
 - ▶ $N=6009$, $\text{age} \in \{33, \dots, 37\}$, and 35% of respondents graduated college
 - ▶ 24% have at least one college-graduate parent

	Mean	Minimum	Median	Maximum
Female	0.527	0	1.0	1
Black	0.270	0	0.0	1
Hispanic	0.210	0	0.0	1
White	0.511	0	1.0	1
Employed	0.757	0	1.0	1
Wage	25.531	8.0	20.0	150.0
College Grad	0.350	0	0.0	1
Age	34.967	33	35.0	37
Parent College	0.239	0	0.0	1
Numkids	1.327	0	1.0	9
EFC	4.224	0.0	0.778	118.111

EXAMPLE: SETTING UP THE SPECIFICATION

- ▶ It looks like we can estimate some form of our model
 - ▶ We have family background, cost of college (EFC variable)
 - ▶ We have employment status, wage and number of children
 - ▶ It looks like we'll have to have s be binary (College grad variable)
 - ▶ Also need to assume $x = age - 18$ if non-grad, $x = age - 22$ if grad
- ▶ Then we just need to add some functional form assumptions, and we'll be ready
 - ▶ $\varepsilon \sim \text{Normal}$, $\eta_t \sim \text{Logistic}$
 - ▶ $u_{i1} = \alpha_0 + \alpha_1 \text{parent college} + \alpha_2 \text{efc} + \eta_1$
 - ▶ $u_{i2} = \gamma_0 + \gamma_1 \mathbb{E} \log w_i + \gamma_2 \text{numkids} + \eta_2$

PARAMETERS OF THE EMPIRICAL MODEL

- ▶ We can now detail the parameters of the empirical model
 - ▶ wage parameters: $(\beta, \sigma_\varepsilon)$
 - ▶ schooling parameters: (α)
 - ▶ employment parameters: (γ, δ)
- ▶ Then write down a statistical objective function as a fn. of data and parameters
 - ▶ e.g. maximize the likelihood, or minimize the sum of the squared residuals
- ▶ We'll learn how to do this in later classes, but not today

3. SOLVING AND 4. UNDERSTANDING HOW THE MODEL WORKS

- ▶ Solving the model:
 - ▶ solve the dynamic utility max problem for given parameter values
 - ▶ (we aren't estimating parameter values yet)
 - ▶ (we will talk about how to do this next week)
- ▶ Understanding the model:
 - ▶ simulate data from the model
 - ▶ make sure the simulated data is consistent with the model's implications
 - ▶ look at descriptive statistics from the simulated data

3. SOLVING AND 4. UNDERSTANDING HOW THE MODEL WORKS

- ▶ More generally, start with as simple of a model as possible; make sure things are working
- ▶ When introducing more complexities, do “numerical comparative statics”
- ▶ Make sure the parameters move in the correct directions
 - ▶ e.g. $\uparrow \beta_1 \implies \uparrow \text{schooling}$ (ceteris paribus)
- ▶ If they don't, you've likely got a bug somewhere
- ▶ Building, simulating, and testing your model in this way will help you:
 - ▶ avoid coding bugs prior to estimation
 - ▶ build intuition for how various parameters impact behavior
 - ▶ provide a simulated data set to test estimation code

5. ESTIMATION

- ▶ Most structural models require **nonlinear estimation**
 - ▶ e.g. MLE/GMM or their simulated counterparts
- ▶ In nonlinear optimization, starting values are crucial
- ▶ Initializing at random starting values is likely to give poor results
- ▶ Keane recommends calibrating the model by hand
 - ▶ e.g. match the intercept of each equation to the $\bar{Y}$'s in the data
 - ▶ you can also estimate an intercepts-only model (or with very few X 's)
- ▶ We will spend lots of time over the next couple of weeks talking about nonlinear estimation

EXAMPLE USING REAL DATA

- ▶ In our simple model, we can get good starting values by estimating OLS and logits
- ▶ The wage equation can be estimated by OLS (on the subsample who are employed)
- ▶ Using stata: `reg lwage collgrad exper exper2`

	Estimate	SE
(Intercept)	2.94607	0.323145
collgrad	0.534326	0.0271395
exper	-0.0265561	0.0412115
exper ²	0.0014304	0.00132307

EXAMPLE USING REAL DATA

- The u_t equations can be estimated as simple logits (on the full sample)

Outcome - College Grad	Estimate	SE
(Intercept)	-1.20091	0.0364888
parent college	1.47866	0.068433
efc	0.0450253	0.00437704

Outcome - Employed	Estimate	SE
(Intercept)	-4.25036	0.454826
Expected log wage	1.80081	0.149078
numkids	-0.0797204	0.0218106

- Then use these as starting values for joint model

DO THESE RESULTS MAKE SENSE?

- ▶ It can be informative to try and interpret even these simple results
- ▶ wage equation:
 - ▶ insignificant return to experience is surprising; otherwise makes sense
- ▶ schooling choice:
 - ▶ If 'efc' captures college costs, it should have a negative sign
 - ▶ This suggests omitted variable bias in this equation
- ▶ employment choice:
 - ▶ These results check out; may want to introduce nonlinearities in 'numkids'

6. VALIDATION

- ▶ If you have a good model, it should be valid (i.e. predict well out of sample)
- ▶ Validation is not always possible, but it's good to do if you can
 - ▶ e.g. if experimental data, estimate on control group, validate on treatment group
 - ▶ e.g. see if model can replicate major policy change in data
- ▶ More simply, you could throw out half your data, then try to predict other half
- ▶ This is typically not done if the full sample isn't huge

7. POLICY EXPERIMENTS

- ▶ This is the main reason to do structural estimation!
- ▶ Structural estimation $\implies$ recovering the DGP of the model
- ▶ Once we know the DGP, we can simulate data from it and do policy experiments
 - ▶ requires having policy-invariant parameters!
- ▶ We can predict the effects of:
 - ▶ proposed policies
 - ▶ hypothetical policies
- ▶ Contrast with RCTs, which only reveal effects of implemented policies

EXAMPLE USING REAL DATA

- ▶ We have two policy variables we could play with
 1. *efc* (i.e. how much gov't subsidizes college tuition & fees)
 2. return to schooling (this could change due to technological change)

- ▶ Since we don't have real estimates from the dynamic model, can't actually do this, but...
 - ▶ simulate behavior in status quo and under various alternative choices for *efc*
 - ▶ simulate behavior for various returns to schooling parameters
 - ▶ this will change schooling choice as agents are forward looking

- ▶ in a few weeks we will talk about how to solve and program these dynamic models

IN SUMMARY: WHY STRUCTURAL ESTIMATION?

- ▶ Want to examine effects of policies not yet implemented
- ▶ Learn more about economics by looking through the lens of a model
- ▶ Assess performance of theoretical models in explaining real-world data
- ▶ Can be used to build up long-run "canonical" models of behavior in many areas
- ▶ It can be really fun to do more complicated econometrics beyond simple regressions
- ▶ Observational data is much cheaper to collect than experimental data

IN SUMMARY: WHY NOT STRUCTURAL ESTIMATION?

- ▶ It's really difficult to write down and estimate a tractable, realistic model!
- ▶ It requires additional effort beyond data preparation and running regressions
- ▶ Understanding identification of the model takes a lot of effort, too
- ▶ It can be really miserable to try and debug the code of a structural estimation
- ▶ Many structural models can take weeks to estimate one specification
 - ▶ in addition to months spent coding/debugging beforehand
- ▶ As you can see, even with a simple model things have already gotten complicated!